Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) building AI/deep-learning applications and regulated software in Python, C#, and JavaScript/TypeScript. Current research applies deep learning and vision-language models to automated analysis of visual/video data; production experience with unit testing, validation, and cloud deployment (AWS, Azure).

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Deep Learning, Advanced Algorithms, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Machine Learning, Data Mining, Data Analytics, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, C#, JavaScript, TypeScript, Java, C/C++, SQL.

AI & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, Gemini, Vertex AI, image/video analysis, model evaluation.

Software Lifecycle: Unit testing, test-script authoring, software validation, SDLC in regulated environments, technical documentation.

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Git, GitLab CI/CD, Jenkins, Terraform, Linux.

Web & Data: React, Node.js, Flask, FastAPI, REST APIs, PostgreSQL, MySQL, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented deep-learning pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity, running full evaluation and validation cycle in Python.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing and resolving live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** through structured design, testing, and rollback validation.
- Served as Quality Coach and SonarQube Specialist, increasing automated unit-test coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Developed **Go + Terratest** unit-test framework, catching **95%** of misconfigurations pre-production, and automated AWS provisioning with Terraform, reducing deployment time by **30%**.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Built full-stack healthcare app with **Vitest** unit/integration test coverage and **Zod** schema validation at every data boundary, eliminating a class of runtime type-mismatch bugs.

News Headline Classification [Python, PyTorch, spaCy] Mar 2025

- Built and unit-tested LSTM-based deep learning classifier on **400K+** news headlines; achieved **93.28% accuracy** outperforming baseline models, and optimized preprocessing pipeline improving training efficiency **25%**.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).